

Neural Networks Recap and Applications

Artificial Intelligence & Machine Learning · Week 2, Day 1

Look around: this room is full of future...

Computer Engineering

Electrical Engineering

Chemical Engineering

Materials Science

Environmental Science

Mechanical Engineering

Computer Science

Today: How neural networks already work inside YOUR field.

Quick recap

Before the cool stuff: what is a neural network?

Every ML system = 4 simple components

- **Data:** the examples we learn from (photos, numbers, text...)

Every ML system = 4 simple components

- **Data:** the examples we learn from (photos, numbers, text...)
- **Model:** the neural network (the AI technique) that makes the prediction

Every ML system = 4 simple components

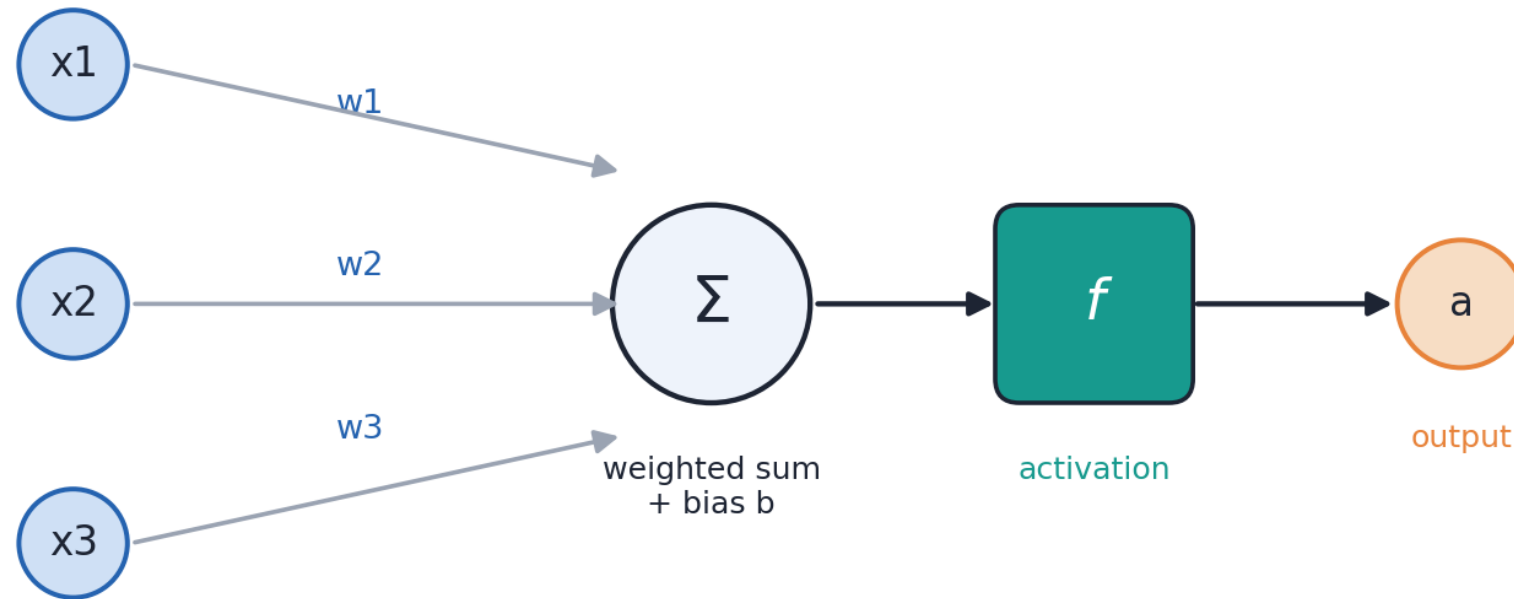
- **Data:** the examples we learn from (photos, numbers, text...)
- **Model:** the neural network (the AI technique) that makes the prediction
- **Loss:** a score of how wrong the prediction is

Every ML system = 4 simple components

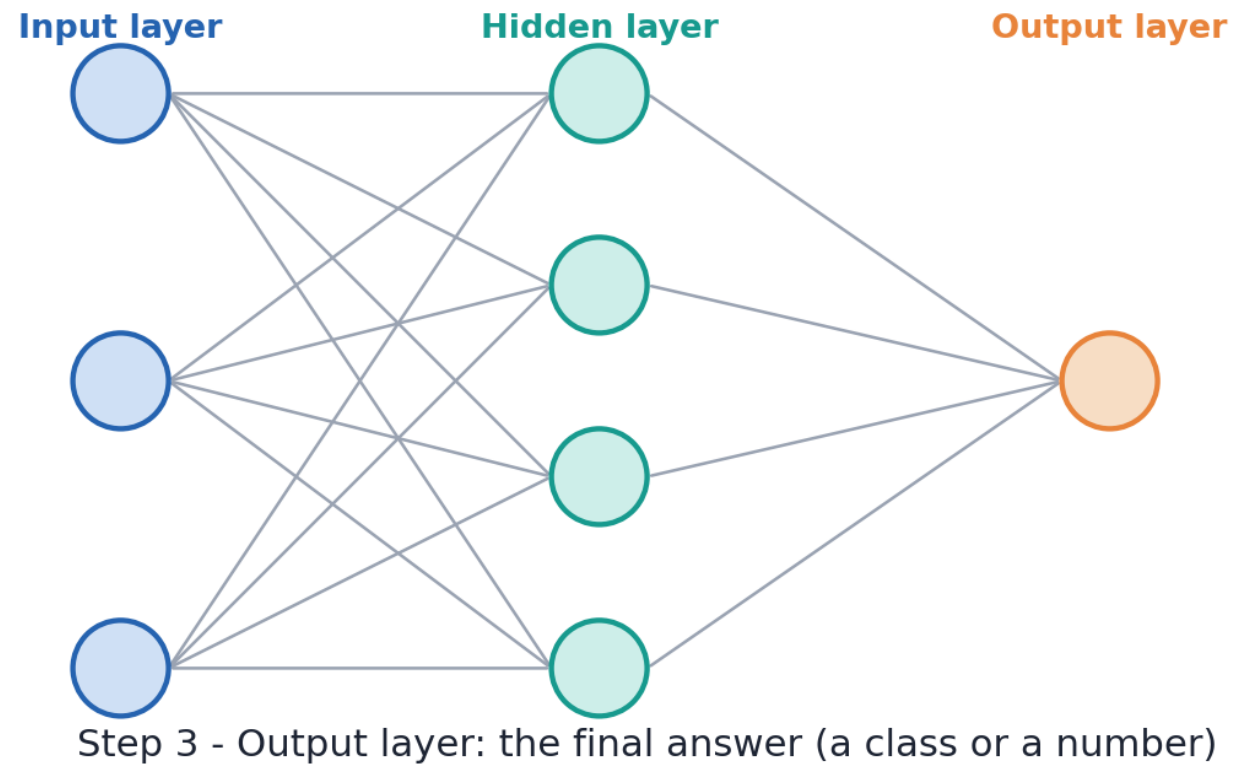
- **Data:** the examples we learn from (photos, numbers, text...)
- **Model:** the neural network (the AI technique) that makes the prediction
- **Loss:** a score of how wrong the prediction is
- **Optimizer:** the rule that improves the model

A neuron: multiply, add, then squash

A single neuron: multiply, add, then squash



Stack neurons $\rightarrow$ a network



Input $\rightarrow$ hidden layers $\rightarrow$ output. 'Deep' = many hidden layers.

QUIZ 1

In a machine-learning system, which one is the MODEL?



The training photos



The neural network making predictions



The error score



The learning-rate setting

Pick your answer!

QUIZ 1

In a machine-learning system, which one is the **MODEL**?



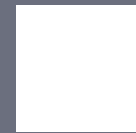
The training photos



The neural network making predictions



The error score



The learning-rate setting

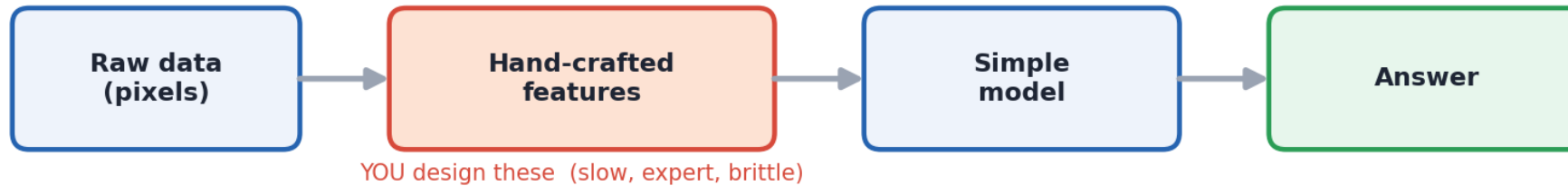
Answer: the Model is the thing that turns inputs into predictions. Data, loss and optimizer are the other three pieces.

First: why not just use traditional ML?

What makes real-world data so hard?

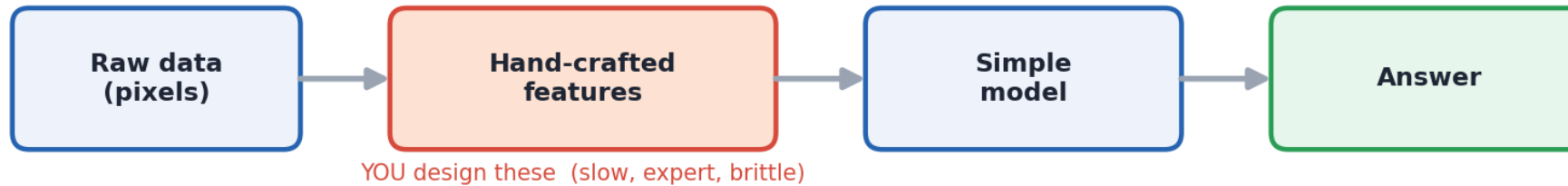
How traditional machine learning works

Traditional ML

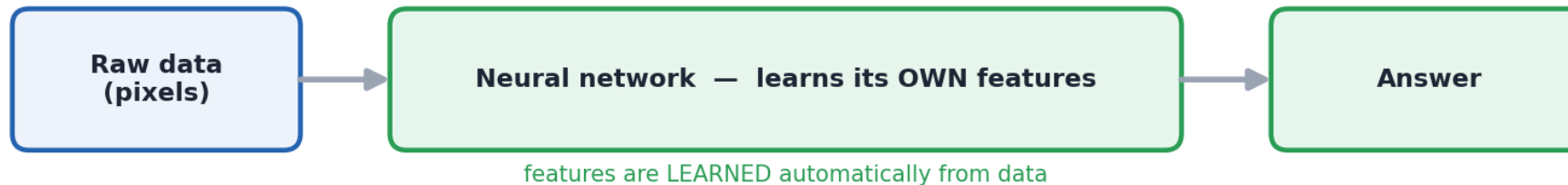


How traditional machine learning works

Traditional ML



Deep Learning



The catch: a human must hand-design the features in the middle.

Try it: write a rule to spot a cat in an image

- **Pointy ears?** dogs, foxes and rabbits have them too
- **Whiskers?** hard to see in a blurry, dark, or side-on photo
- **Fur colour?** cats come in every colour and pattern
- **A tail?** so do most animals - and it might be hidden
- By hand this is almost impossible: too many exceptions to write down.

Structured vs unstructured data

Structured (easy)

Neat rows and columns - spreadsheets, sensor logs, prices.
Traditional ML is great here.

Unstructured (hard)

No tidy columns - images, audio, text, video. Most real-world data lives here!

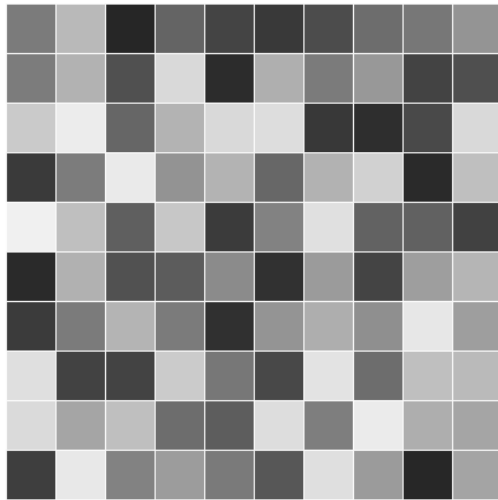
The world is mostly UNSTRUCTURED - exactly where traditional ML struggles

So what makes data 'complex'?

- 1 **High-dimensional:** one photo = thousands of pixel numbers
- 2 **Raw & unstructured:** pixels, soundwaves and text, not tidy columns
- 3 **Non-linear:** the real pattern is curvy, not a straight line
- 4 **Hugely varied :** the same cat in different poses, lighting and angles
- 5 **Massive scale:** millions of examples needed to learn from

Complexity 1 - high dimensionality

a tiny 10×10 photo



0.21
0.94
0.07
0.55
...
0.83
0.12

= a list of
100 numbers

A normal 100×100 photo = 10,000 numbers. A phone photo = millions.



More dimensions = the 'curse of dimensionality'

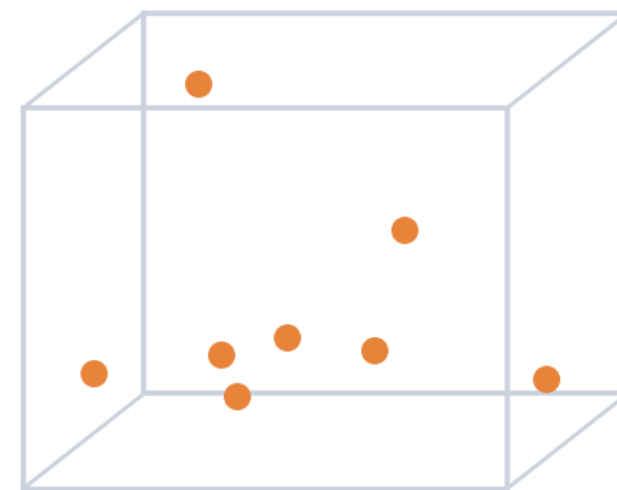
1D: points packed together



2D: same points, more spread



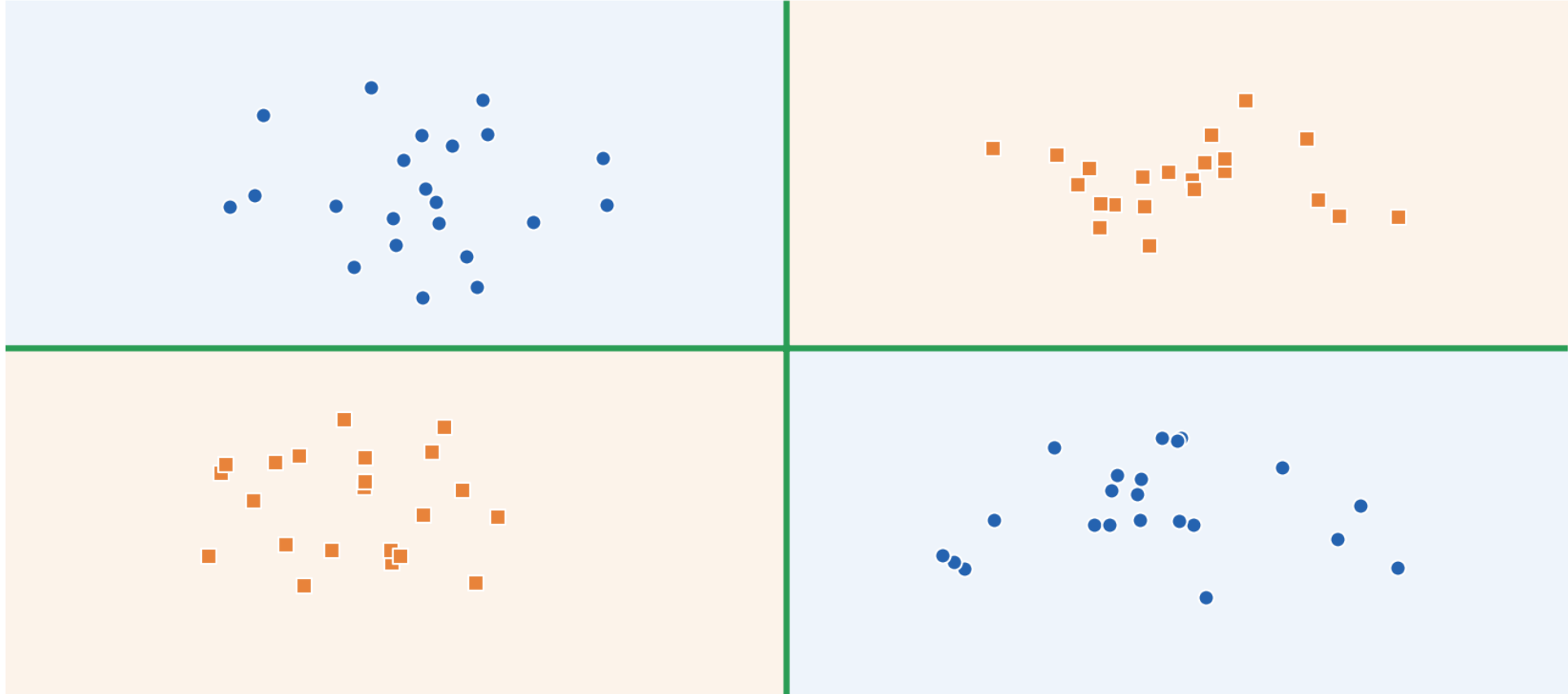
3D: points very far apart



The 'curse of dimensionality': more features → data spreads thin → models need far more examples

More features -> examples spread thin -> simple models need far more data.

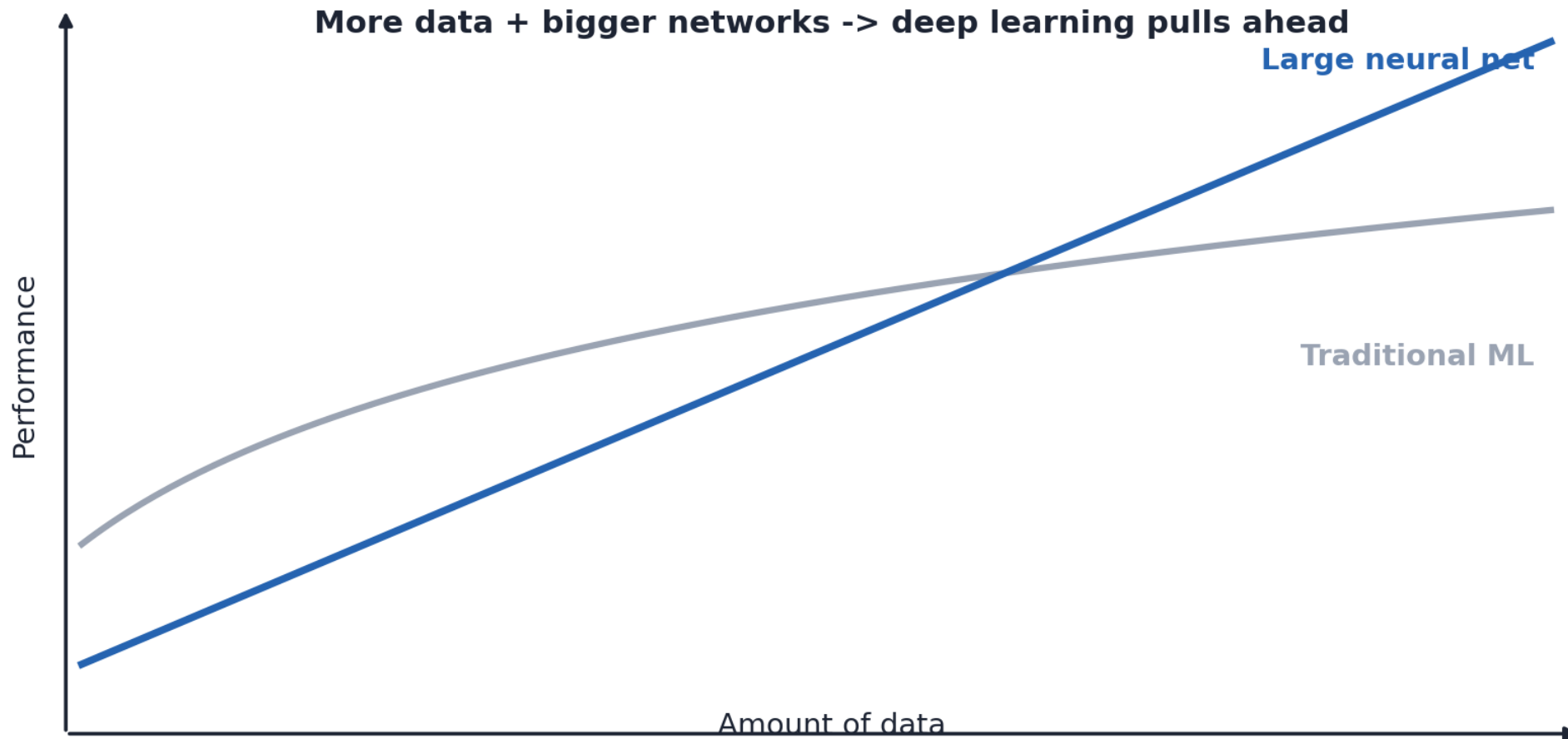
Complexity 2 - patterns are non-linear



A curved boundary separates them perfectly.

A straight line can't separate this; real data needs curved boundaries.

So traditional ML hits a wall



Even with more data, hand-built models has lower performance but flexible deep neural nets keep improving.

QUIZ 2

Why is it hard for traditional ML to recognise a cat directly from raw pixels?



Pixels are too colourful



Someone must hand-design features - nearly impossible



Cats have no features



Computers can't open images

Pick your answer!

QUIZ 2

Why is it hard for traditional ML to recognise a cat directly from raw pixels?



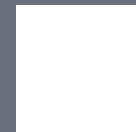
Pixels are too colourful



Someone must hand-design features - nearly impossible



Cats have no features



Computers can't open images

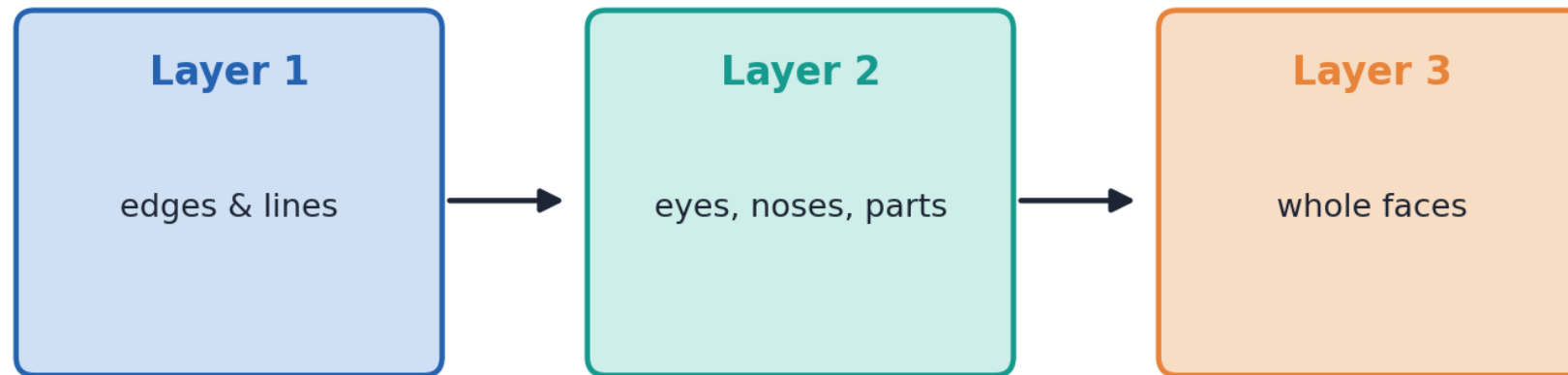
Answer: raw pixels carry no built-in meaning, and hand-designing features for every case is impractical.

Deep learning's big idea

Let the network LEARN the features by itself

Layer by layer: simple -> complex

Deep networks build understanding in layers



Simple features combine into complex ones - the network is not told what to look for.

Early layers find edges; deeper layers combine them into objects.

The bottom line

- **Traditional ML** needs humans to hand-craft features and breaks on complex data
- **Complex data** high-dimensional, unstructured, non-linear, varied and huge
- **Deep learning** learns the right features automatically, and improves with scale
- **So...** that's why neural networks power the tools in YOUR field, let's see how

Neural Networks in YOUR Field

Same idea, wildly different jobs/objectives

Computer & Electrical Engineering

1

Vision

Face unlock, object & defect
detection from camera images

2

Voice

Speech assistants (Siri, Alexa) turn
audio into commands

3

Hardware

Chip layout design & predicting
power-grid demand

Data types: images · audio · signals

Chemical Engineering

1

Molecules

Predict a molecule's properties before making it

2

Processes

Optimize reaction yield & control plants in real time

3

Drugs

Screen millions of candidate compounds fast

Data types: molecular structures · sensor tables

QUIZ 3

Your phone unlocking by recognizing your face uses a neural network. What kind of data is it working with?



Plain text



Audio clips



Images (a grid of pixels)



Spreadsheet rows

Pick your answer!

QUIZ 3

Your phone unlocking by recognizing your face uses a neural network. What kind of data is it working with?



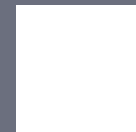
Plain text



Audio clips



Images (a grid of pixels)



Spreadsheet rows

Answer: a photo is just a grid of pixel numbers i.e., image data. (We'll see why images are tricky next class!)

Materials Science & Engineering

1

Discovery

Find new battery materials & alloys
without testing everyone

2

Properties

Predict strength or conductivity from
structure

3

Imaging

Spot defects in microscope images
automatically

Data types: crystal structures · microscopy images

Environmental Science

1

Climate

Forecast weather & climate patterns

2

Air & water

Predict air quality from sensor networks

3

Satellites

Detect deforestation & wildfires from space imagery

Data types: sensor time-series · satellite images

QUIZ 4

Scientists use neural nets to discover new battery materials. Why use AI instead of testing every material in the lab?



AI is always 100% correct



Testing every option would be far too slow & costly



Labs are closed



Materials don't matter

Pick your answer!

QUIZ 4

Scientists use neural nets to discover new battery materials. Why use AI instead of testing every material in the lab?



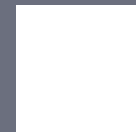
AI is always 100% correct



Testing every option would be far too slow & costly



Labs are closed



Materials don't matter

Answer: there are millions of possibilities. AI predicts the most promising few, so labs only test those.

Mechanical Engineering

1

Maintenance

'Hear' a failing motor from vibration
before it breaks

2

Robotics

Control robots and self-driving
systems

3

Simulation

Instant fluid/stress estimates instead
of slow physics runs

Data types: vibration signals · sensor streams

Computer Science

1

Language

ChatGPT & LLMs predict the next word in a sentence

2

Recommendations

YouTube / Netflix / TikTok suggestions

3

Generation

Image generators like DALL·E and Midjourney

Data types: text · clicks · images

QUIZ 5

What makes ChatGPT able to 'understand' and answer your sentence?



It stores every sentence ever written



A neural network trained on huge amounts of text



A human types each reply



Pure magic

Pick your answer!

QUIZ 5

What makes ChatGPT able to 'understand' and answer your sentence?



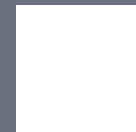
It stores every sentence ever written



A neural network trained on huge amounts of text



A human types each reply



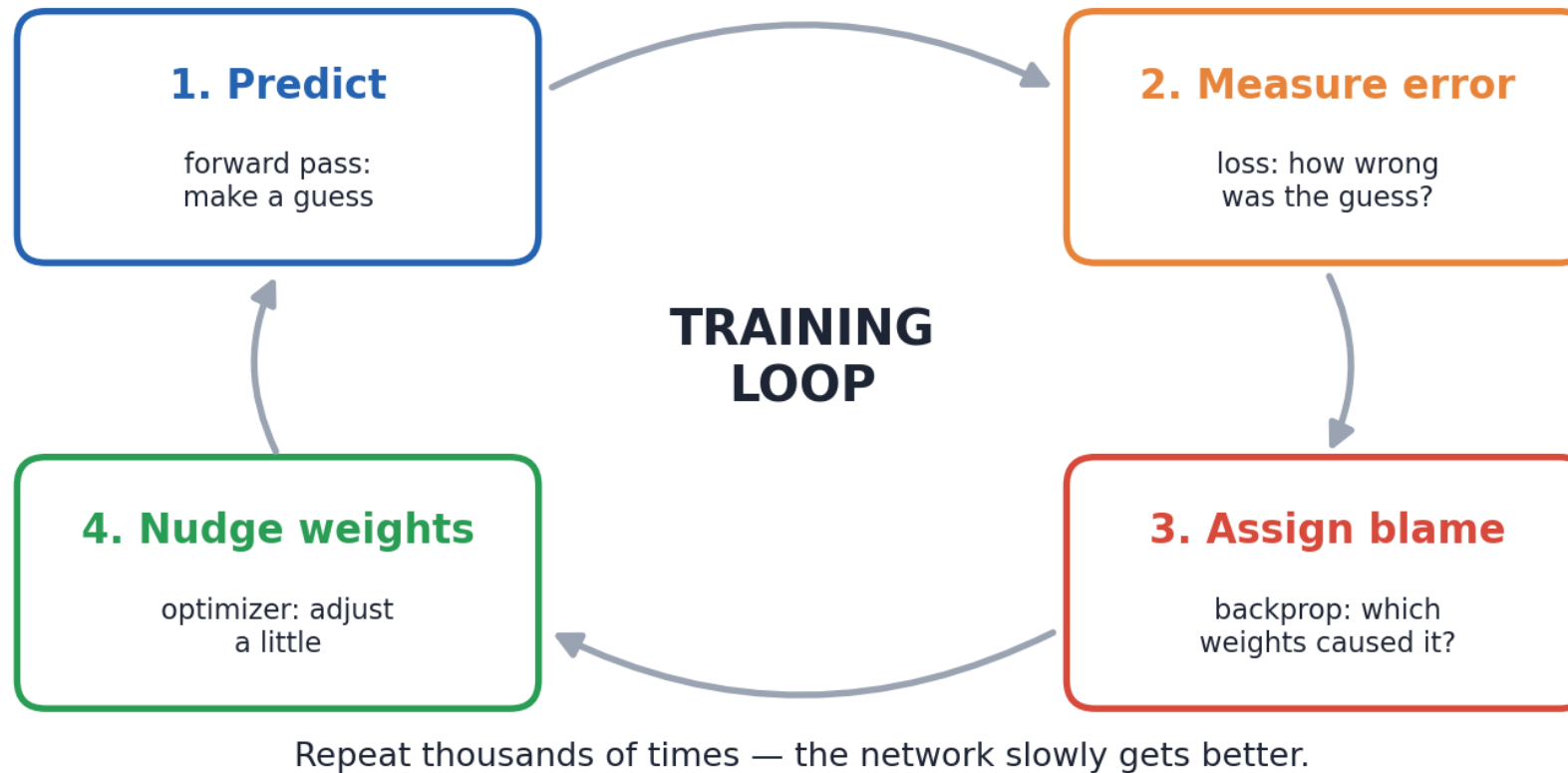
Pure magic

Answer: large language models are giant neural networks trained to predict the next word, over and over.

But how do they actually **LEARN**?

Training = predict, get feedback, adjust and then repeat

The training loop



Round and round: every loop makes the network a little less wrong.

QUIZ 6

During training, what does the network actually change to get better?



The training data



Its weights: the numbers inside



Your question



Nothing: it's fixed

Pick your answer!

QUIZ 6

During training, what does the network actually change to get better?



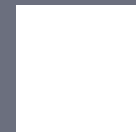
The training data



Its weights: the numbers inside



Your question

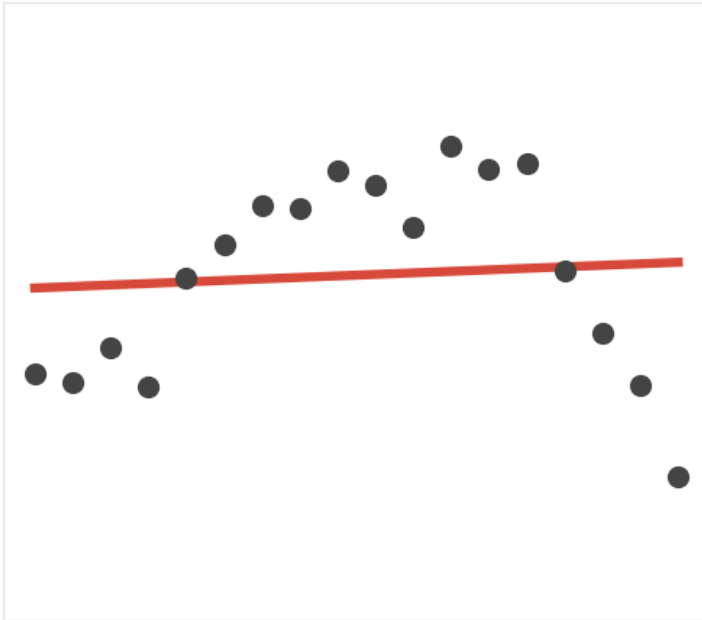


Nothing: it's fixed

Answer: learning literally means adjusting the weights to shrink the loss, step by step.

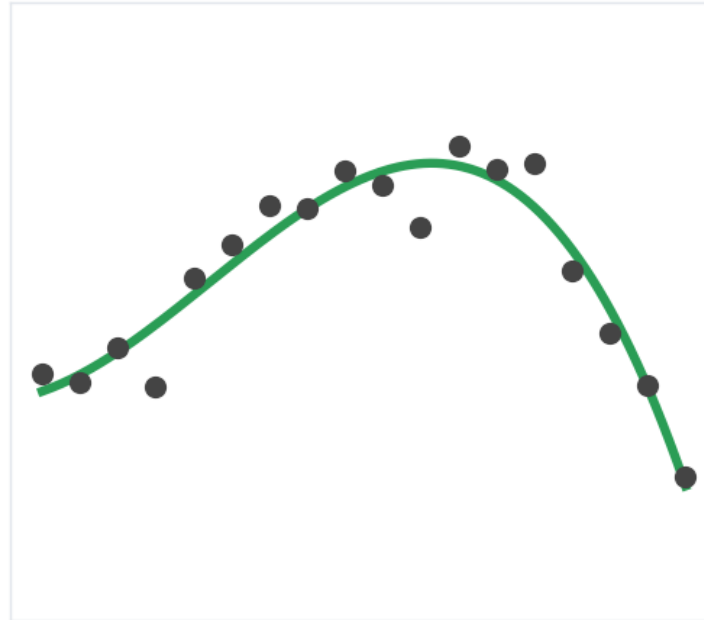
Learning, not memorizing

Underfitting



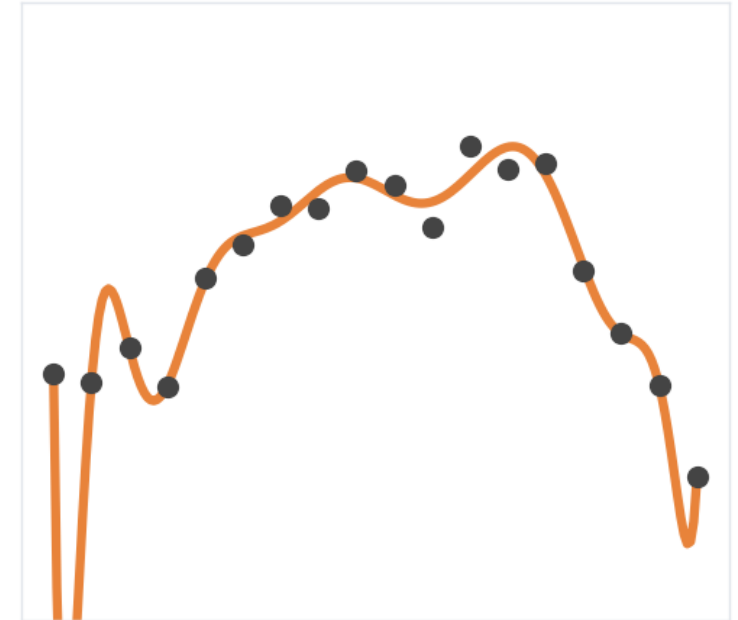
too simple — misses the pattern

Good fit



learns the real trend

Overfitting



memorizes the noise

We hold back some data to check it truly learned: not just memorized.

Different problems, different answers

Regression

Predict a NUMBER

e.g. material strength,
tomorrow's temperature,
house price

Yes / No

A TRUE/FALSE choice

e.g. defective part?
spam email?
tumour present?

Pick one of many

Choose a CATEGORY

e.g. digit 0–9,
which species,
which language

Same network idea: the **OUTPUT** changes: one number · a yes/no probability · the top category

QUIZ 7

Predicting tomorrow's temperature (a single number) is which kind of problem?



Classification



Regression



Clustering



Translation

Pick your answer!

QUIZ 7

Predicting tomorrow's temperature (a single number) is which kind of problem?



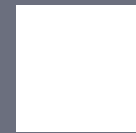
Classification



Regression



Clustering



Translation

Answer: predicting a number = regression. Sorting things into categories = classification.

Same toolkit, many real jobs

- **Classify** sort emails, images, or molecules into categories
- **Predict a number** forecast demand, temperature, or material strength
- **Detect & locate** find objects or defects inside an image (more on Day 4!)
- **Generate** create new text and images: ChatGPT, DALL-E
- **Recommend** suggest the next video, product, or paper

Different data needs different networks

Tables of numbers

Plain neural net (MLP)

e.g. engineering data, prices, sensors

Images

CNN › next class!

e.g. photos, scans, satellites

Text & sequences

RNN / Transformer

e.g. language, audio, time-series

Same neuron idea but arranged differently for each kind of data.

QUIZ 8

We add 'activation functions' to a neural network so it can...



Run faster



Learn curved patterns, not just straight lines



Use less data



Skip training entirely

Pick your answer!

QUIZ 8

We add 'activation functions' to a neural network so it can...



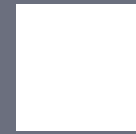
Run faster



Learn curved patterns, not just straight lines



Use less data



Skip training entirely

Answer: without activations, stacking layers stays a straight line. Activations add the bends that fit real data.

Key takeaways

- A neural network is just a flexible Model built from simple neurons
- The same idea powers tools across every field in this room
- Different data (tables, images, text) → different network designs
- Coming classes: why images break a plain neural network and how CNNs fix it



Questions?

Next time: when images break neural networks